RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Even Semester)

<u>Innovative Practices Description</u>

<u>Unit IV (MAGNETOSTATICS)</u>

Degree, Semester& Branch: IV Semester B.E. ECE B
Course Code & Title: EC8451 Electromagnetic Fields
Name of the Faculty member: Mr.P.Venkatesh
Name of the Topic: Magneto statics
Name of the Innovative Practice: Think-pair-share
Date: 04.02.2020

Think-pair-share:

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to (1) think individually about a topic or answer to a question; and (2) share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material.

<u>Goals (Learning Outcomes):</u>

The students will be able to

1. Analyse the magneto static behavior of materials
2. Analyze the issues magneto statics.

<u>Use of appropriate methods:</u>

Reason for choosing the activity for the topic:

Magneto statics have many steps in designing and classifications which require following proper order according to the given statement. As the activity involves various aspects of learning and also maps out versatile ideas, this activity was chosen.

<u>Effective presentation:</u>

Implementation (Plan & Execution) with Proof:

The activity was planned for duration of 10 minutes. First the students were asked to think about the topic given. Then they were paired among their nearby students. They should then share the ideas that they have recollected and list it down in a paper. Later some among the students (pair) were asked to come front and share their ideas that they have written with the entire class.

Innovative Teaching Method Execution
Introduction to asynchronous circuits – Think-pair-share

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

This strategy in developing critical thinking skills among the students. Think Pair Share is one of the learning strategies that encourages student to participate in group discussion.

Reflective Critique:

Benefits:

i) The personal interaction motivates students who might not generally be interested in the discipline.

ii) It engages the entire class and allows quiet students to answer questions without having to stand out from their classmates.

iii) Recollection of ideas will be easier as they are combined as pairs

Challenges:

i) One of the biggest challenges of the think-pair-share is to get all students to truly be engaged.

ii) That is students among different pairs may share the same ideas. In such case instructor may ensure the genuineness.

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO3	2	2	2	1				2	2	3		

CO3: The student will be able to analyze the field potentials due to static magnetic fields and explain how materials affect electric and magnetic fields

References:

i) D.K. Cheng, Field and wave electromagnetics, 2nd ed., Pearson (India), 1989 (UNIT I, II,III IV,V)

ii) W.H. Hayt and J.A. Buck, Engineering electrmagnetics, 7th ed., McGraw-Hill (India), 2006 (UNIT I-V)